PI3K-mini: A Positive and a Negative Result for FIM Null-Space Regularization

What a small, fully-controlled toy model taught us, and what it predicts for the real HGF model

Companion to `scripts/example_toy_pi3kmini_fim_fit.R`, `scripts/example_toy_pi3kmini_fim_plots.R`, and `scripts/example_toy_pi3kmini_fim_eigenvalue_report.R`

1 Purpose

This note records the reasoning, experiments, and conclusions of an extended investigation into whether the `groupL2()` / `nullSpaceProjector()` FIM null-space regularization (`fim-portugal/functions-port` can be made to work reliably, using a small, fully-transparent toy model built specifically to test the method's own precondition under full control. The model, "PI3K-mini", is a deliberate mechanistic miniature of the PI3K structural non-identifiability already documented for the real HGF/HepG2 model in `HGF_HepG2/04-Results/M-22.0-comparison/05-Writeup/Technical-Notes_PI3K-Symmetry-and-Regularization-Diagnostics.pdf`. The investigation produced both a clean positive result (Section 3) and, on pushing the same model toward more realistic data, a clean negative one (Section 6) — together with a precise, verified mechanistic explanation for the transition between the two (Sections 7–9), a proposed hybrid regularizer that keeps the positive part without inheriting the negative part (Section 11), and an assessment of what this predicts for the real, 96-parameter HGF model (Section 12).

2 The model

2.1 Topology

Reaction	Role
$A \to B$, rate $k_1 A$	control block; directly observed; unrelated to the symmetry
$\varnothing \to U$, rate $k_{\text{gen}} A$	generates hidden U catalytically (A not consumed)
$U \to \varnothing$, rate $k_{\text{degU}} U$	U's own decay; *not* part of the symmetry
$M_p \to M_{pp}$, rate $k_{\text{use}} M_p U$	only readout of U's activity, via obsM_{pp}

Three conditions (`cond1/cond2/cond3`) differ only in U's initial value ($U_{\text{cond1}} = 1$, $U_{\text{cond2}} = 3$, $U_{\text{cond3}} = 6$), mirroring `PI3K_5/PI3K_250/PI3K_350` in the real model. $A_0 = 1$ and $M_{p,0} = 1$ are fixed (not fitted) to avoid two *unrelated* accidental scale symmetries ($A_0 \leftrightarrow \text{scale}_A/\text{scale}_B$ and $M_{p,0} \leftrightarrow \text{scale}_{Mpp}$) that this design would otherwise introduce.

2.2 The exact symmetry

For any $c > 0$:

$$U_{\text{cond}i} \to c\,U_{\text{cond}i}\ (i = 1, 2, 3), \qquad k_{\text{gen}} \to c\,k_{\text{gen}}, \qquad k_{\text{use}} \to k_{\text{use}}/c$$

leaves $U(t)$ scaling exactly by c (linear, homogeneous ODE in U) and leaves $k_{\text{use}}U(t)$ — the only combination M_p/M_{pp} ever see — exactly invariant. Every observable trajectory is therefore exactly invariant under this one-parameter family, verified numerically to solver precision ($\sim 10^{-7}$, Section 3). k_{degU} does *not* load on this symmetry (linear decay of U needs no compensation), mirroring why the real model's `k_degrade_PI3K` does not load on the real PI3K symmetry either.

The 5-parameter group $(U_{\text{cond1}}, U_{\text{cond2}}, U_{\text{cond3}}, k_{\text{gen}}, k_{\text{use}})$ spans a rank-1 null space (one non-identifiable direction) and a rank-4 identifiable complement, spanned by the four combinations $k_{\text{use}}U_{\text{cond1}}$, $k_{\text{use}}U_{\text{cond2}}$, $k_{\text{use}}U_{\text{cond3}}$, $k_{\text{gen}}k_{\text{use}}$ — the same 1-null-in-5 structure as the real PI3K group. This was a deliberate design choice: the minimal 2-parameter AB-model ($k \cdot q$, §5.2 of the thesis draft) has *no* identifiable complement at all (rank-1 null space inside a 2-dimensional group exhausts the space), so it cannot show the distinction between "the non-identifiable direction" and "a raw parameter that happens to participate in it" that turns out to be central to this whole investigation (Section 9).

2.3 An implementation pitfall

`Y(..., attach.input = TRUE)` attaches *all* raw states, including the hidden U, to the prediction output. Building the simulated dataset from that attached output without filtering down to the actual observables first silently fits against U's own (non-invariant) trajectory too, which destroys the intended symmetry with no error or warning — the Hessian's eigenvalue spectrum simply looks like an ordinary sloppy spread, no gap anywhere. `attach.input = FALSE` is required. (This cost real debugging time: the first version of the fit showed no spectral gap at all, which briefly looked like the toy model design itself was wrong before the true cause was found by directly comparing predicted trajectories under a hand-applied symmetry shift against what `normL2()` was actually being handed as "data".)

3 Result 1: a clean win, on dense/precise data

With $\sigma = 0.03$, 31 time points per condition (dense, precise data), 200 multistart fits per objective (`obj_data`, `obj_l2 = obj_data + constraintL2(...)`, `obj_fim = obj_data + groupL2(P_null,...)`), all three centered at the *same* reference point $\mu = -1$ in every log-parameter (a fair-comparison fix — see Section 4):

	no regularization	FIM (group)	L2 (uniform)
raw data χ^2 at best fit	−1186.38	−1186.38	−1186.28
identified-block RMSE	0.099	0.099	0.159
invariant-combo RMSE	0.061	0.061	0.081
SD of raw U_{cond1}, near-optimal fits	2.06–7.6	$\approx 10^{-4}$	$\approx 10^{-4}$

FIM matches the unregularized fit's accuracy exactly (unbiased) while matching L2's reproducibility exactly (bounded) — strictly better than either alternative on its own, in the regime where the method's own precondition (one isolated exact symmetry, everything else genuinely sharp) holds

by construction. Eigenvalue spectrum (log10) at this data quality: 5.26, 5.16, 4.51, 3.90, 3.58, 2.94, 2.68, 2.04, 0.5
— one clean ∼12-order-of-magnitude gap, nothing else close to it.

4 Same regularization center for both methods

An earlier version anchored `groupL2()` at `best_unreg` (the single best unregularized fit). Along the exactly flat direction this anchor is itself an arbitrary draw from the unregularized multistart's own spread — in one run it landed at $\log U_{\text{cond1}} = 2.84$ (true value 0), so all FIM-regularized fits collapsed to that arbitrary point instead of anywhere near truth, while L2 (anchored at $\mu = 0$, coincidentally close to U_{cond1}'s true value 1) looked "better" for a reason that had nothing to do with either method's merit. Fix: anchor both penalties at the *same* $\mu_{\text{center}} = -1$ for every parameter. This is mathematically required for a fair comparison and also exposed a useful fact: with a shared center, L2 and FIM land at (numerically) the *same* point along the null direction, because L2's cost on the stiff/identified directions is constant along that valley — so minimizing L2 restricted to the valley reduces to the identical 1-D problem FIM's rank-1 penalty solves directly.

Caveat worth remembering: the individual raw value a regularizer settles on along a truly non-identifiable direction is not itself a meaningful quantity — it is set entirely by the chosen anchor, not by data. Comparing raw per-parameter estimates between two differently-anchored regularizers is not a fair test of anything.

5 Two outlier local optima under FIM

At the dense-data setting, dMod's own `plotPars()` (step-detected via `stepDetect()`) flags 2 of 200 FIM fits as genuinely worse: raw (unpenalized) $\chi^2 = -1086.1$ and -729.4 vs. -1186.4 for the rest — 100 and 457 worse, both reported `converged=TRUE`, 29–38 iterations, not truncated runs. The FIM penalty contributes ≈ 0 at all three points, confirming these are genuine alternative local optima of the raw likelihood, not penalty artifacts. Predicted curves (`pi3kmini_fim_outliers.png`) show obsM_{pp} in `cond2`/`cond3` saturating almost instantly instead of following the data's gradual early rise — a classic "which rate governs the fast vs. slow phase" ambiguity in fitting a hidden-species-driven rise, unrelated to the exact symmetry (parameter shifts from the best fit are *not* consistent with a pure c-rescaling: U_{cond3} moves independently of $U_{\text{cond1}}/U_{\text{cond2}}/k_{\text{gen}}$). Neither unregularized nor L2 hit this basin in the same 200-start sample (0/200 each); most simply attributed to a small ($\sim$1%) basin and ordinary multistart sampling luck, not confirmed to be FIM-specific.

6 Result 2: sparser/noisier data reverses the result

Reducing to 14 time points and raising σ (later swept from 0.05 to 1.7 at fixed data density) reverses the earlier ranking:

	no regularization	FIM (group)	L2 (uniform)
identified-block RMSE	1.19	1.19	0.38
invariant-combo RMSE	6.36	5.89	0.33
raw χ^2 at best fit	−191.6	−191.6	−190.9

L2 is now clearly *better* than FIM on exactly the quantities FIM was supposed to protect. The eigenvalue spectrum at 14 points shows a new, second gap that did not exist at 31 points: $\ldots, 0.36, -1.25, -2.44, -13.3$ (log10) — the true symmetry gap is untouched (still $\sim$11 orders), but ranks 8–9 are new, genuinely emergent near-degenerate directions.

7 Diagnosing the new directions

The rank-8 eigenvector ($\lambda \approx 10^{-1.25}$) is *not* the true symmetry: it loads heavily on k_{gen} (0.84) and $U_{\text{cond1/2/3}}$ (-0.28 each) with essentially *no* k_{use} component (-0.003) — a k_{gen}-vs-$U_{\text{cond}i}$ trade-off distinct from the true symmetry, which requires k_{use} to move oppositely. Projecting the near-optimal fits onto this new direction vs. onto the true null direction (both computed from the data's own H) gives a definitive result:

projection onto	unreg SD	FIM SD
true exact symmetry (rank 10)	2.32	1.5×10^{-13}
new emergent direction (rank 8)	2.51	2.19

FIM controls the direction it was built from essentially perfectly, and provides *zero* protection for the new one — confirmed across every saved σ: the true-symmetry-projection SD stays at 10^{-13}–10^{-15} at *every* σ tested, unconditionally (Section 8 explains why this holds exactly, not just approximately).

7.1 Correction: uniform noise does not reshape the spectrum

A uniform σ applied identically to every data point gives $H = S'S/\sigma^2$ — an exact positive scalar multiple of the same matrix at every σ. Eigenvectors of a scalar-multiplied matrix are unchanged; only eigenvalues scale. Checked directly (`pi3kmini_eigenvalue_report.csv`, `pi3kmini_eigenvalues_vs_sigma.pdf`): consecutive-rank log10-gaps are identical to 2 decimal places across every saved σ (0.05–1.7) except the last (true-symmetry) one, which is pure numerical noise around the exact zero. *The "second gap" found in Section 6 came from reducing the number of time points (which changes S itself), not from raising σ (which only rescales H uniformly and provably cannot reshape relative gaps).* P_{null} itself was verified numerically identical (max abs. difference 6×10^{-13}) between $\sigma = 0.05$ and $\sigma = 1.7$.

8 Why noise-driven curvature loss still hurts FIM

Even with P_{null} and the gap structure fixed, raising a uniform σ still shrinks *every* data-only curvature entry as $1/\sigma^2$ — including the four identifiable combinations inside the same 5-parameter group. `constraintL2` adds a curvature of exactly $2/\sigma_{L2}^2$ to *every* coordinate, independent of σ; as data curvature shrinks below this fixed floor, L2 increasingly (and correctly, in the sense of "it always would") dominates every coordinate, including the ones that were never part of the symmetry. `groupL2` adds curvature only along P_{null}'s span — exactly zero elsewhere, at every σ — so the identified complement's precision degrades with σ exactly as it would under no regularization at all. This is not a flaw specific to the implementation; it is the necessary consequence of a penalty that is deliberately inert outside one, exactly-known direction.

9 Does pinning the null direction give unique parameter values?

A natural expectation: if the null direction is regularized, doesn't the whole 5-parameter block become uniquely (hence tightly) estimable? The answer requires separating two different claims.

True: once P_{null} is regularized, the full block has a mathematically unique minimizer for a fixed dataset, because the other four directions have real (if possibly small) curvature — there is no longer an infinite flat family.

Not automatically true: that this unique minimizer is found *precisely* by any real optimizer, or corresponds to a *practically* tight estimate. When curvature on those other directions is small, the objective is nearly flat over a wide region around the unique minimum: a small gradient (near any optimizer's stopping tolerance) corresponds to a *large* parameter displacement, since displacement ≈ gradient / curvature. Different multistart runs land at genuinely different points satisfying the same loose tolerance; even the exact minimizer, if found, would carry a wide, honest confidence interval on that direction.

This is exactly the **structural vs. practical identifiability** distinction of Raue et al. (2009, already cited in the thesis bib). FIM's projector removes only *structural* non-identifiability (an exact, dataset-independent symmetry). It has no mechanism against *practical* non-identifiability (a technically-unique but data-quality-dependent flat region) — and that is precisely what grows in the other four directions as σ increases. No method that targets only a fixed, precomputed exact symmetry can fix that, because it is not a symmetry, only weak information.

9.1 This is not special to PI3K-mini's 5-parameter group

The identical logic applies to the AB-model's k, q: its "identified" direction $(k \cdot q)$ has data-driven curvature that would also shrink under worse data, and k, q would show the same growing, *correlated* scatter (moving together along the weakening product direction, while staying locked along the still-perfectly-regularized k/q direction) that `ab_fit.R`'s own dense/precise dataset simply never demonstrated. The 2-parameter AB-model cannot show the richer phenomenon of Section 7 (a raw parameter mixing a protected direction with unprotected ones), because with only 2 parameters and 1 null direction there is no room for a third, independent direction to emerge — but the core mechanism (a fixed penalty vs. a shrinking data curvature) is identical in both models.

9.2 Summary statement for the chapter

With sparser and noisier data, the model gets more *sloppy* (Gutenkunst et al., 2007): every direction's curvature shrinks, and previously-comfortable directions sink below whatever curvature is needed to be practically useful — not because the model's algebra changed, but because the ability to resolve it with this data did. Sparser sampling (design) can additionally reshape the spectrum, introducing genuinely new near-degenerate combinations; uniform noise alone only rescales the existing spectrum, but that alone is enough to push previously-fine directions below the practical threshold. FIM's projector encodes only the first, permanent kind of fact (a proven exact symmetry); it has no mechanism for the second (data-quality-dependent weakness on directions that are structurally fine).

10 Generalization: how many groups is not the point

A model with two (or more) independent, exactly-isolated symmetry groups, with a genuinely sharp complement, would be handled by FIM exactly as well as one group: `nullSpaceBasis()`'s threshold classification and `nullSpaceProjector()`'s construction are entirely general in the rank of the null space, and nothing in the mechanism cares how many dimensions are exactly zero, only whether they are cleanly separated from everything else. *The number of exact-symmetry groups is not what breaks the method; a gapless, sloppy complement is.* This reframes the difference between PI3K-mini's clean case and the real M-22.0 model precisely: M-22.0 does not fail because it has "more" non-identifiability than a toy model — it fails because, beyond the one clean PI3K degeneracy, the remaining $\sim$91 parameters show no gap at all (a smooth decline over many orders of magnitude). A model with three clean PI3K-like degeneracies and an otherwise sharp, gapped spectrum would likely see FIM succeed just as cleanly as it did here.

11 A hybrid regularizer

Since `groupL2()` and `constraintL2()` act on orthogonal pieces of parameter space, they can be summed:

$$\text{obj_hybrid} = \text{obj_data} + \texttt{groupL2}(P_{\text{null}}, \mu, \lambda_2) + \texttt{constraintL2}(\mu, \sigma_{L2}), \qquad \sigma_{L2} = 1/\sqrt{2\lambda_1}.$$

Equivalently, using $P_{\text{stiff}} := I - P_{\text{null}}$ and two `groupL2` calls with strengths λ_1, λ_2:

$$\lambda_1 P_{\text{stiff}} + \lambda_2 P_{\text{null}} = \lambda_1(I - P_{\text{null}}) + \lambda_2 P_{\text{null}} = \lambda_1 I + (\lambda_2 - \lambda_1) P_{\text{null}}.$$

This is *always* "plain isotropic L2 at strength λ_1, plus a correction confined to the null direction of size $\lambda_2 - \lambda_1$" — the $\lambda_1 I$ term is not conditional on $\lambda_1 = \lambda_2$; it falls directly out of P_{stiff}'s definition as $I - P_{\text{null}}$, for any λ_1. Setting $\lambda_1 = \lambda_2$ collapses the whole construction to plain L2 (isotropic penalties are rotation-invariant: equal weight across *any* complete orthogonal partition of the space reproduces λI, whether that partition is the raw parameter axes or a null/stiff eigenbasis split). Only $\lambda_1 \neq \lambda_2$ produces something genuinely different from plain L2.

Predicted behavior, following directly from Sections 3–9: matches FIM's unbiasedness on the one proven-exact direction (strong λ_2 dominates there); provides L2's boundedness/crash-resistance floor everywhere else (weak λ_1, instead of FIM's exactly-zero protection there), at the cost of the same small, honest L2-style bias on those other directions, scaled down by how small λ_1 is set. This is not a fix for genuine sloppiness — the "everything else" term still pays exactly the same data-blind L2 tax plain L2 always has — it is a more careful *allocation* of where that tax is paid: never on the one direction provable to be exact, always (lightly) elsewhere. A three-or-more-tier version (extra protection specifically for "moderately sloppy but not exactly null" directions, based on their eigenvalue) would re-inherit the *no-free-lunch* problem already documented in `Technical-Notes_PI3K-Symmetry-and-Regularization-Diagnostics.pdf` (§3–4): there is no principled threshold separating "weak but real" from "weak and worthless" in a genuinely gapless spectrum. The two-tier version above is defensible specifically because its boundary sits at an algebraically *proven* zero, not an arbitrary cutoff.

Not yet implemented/tested as of this note; a natural next step (`obj_hybrid` as a fourth objective in the fitting script, rerun across the same σ sweep).

12 Implications for the real HGF model, and chapter strategy

The existing M-22.0 analysis (`Chapter-Outline_FIM-vs-L2-regularization.pdf`) already shows exactly the signature this toy model predicts: one clean, reproducible, mechanistically-explicable exact degeneracy (PI3K) found cleanly by the same method, embedded in an otherwise gapless, $\sim 90/96$- direction sloppy spectrum. The toy model does not predict a *different* outcome for the real model; it explains the outcome the real model already gave, and demonstrates the same transition (clean win $\rightarrow$ failure) happening to one direction at a time, on demand, in a system small enough to eigendecompose and inspect by hand.

A separate, larger, more biologically comprehensive version of the HGF model was also discussed as a candidate addition to the chapter (the version originally requested by the biology collaborators, which needed heavy hand-tuned priors, converged poorly, and crashed frequently before being cut down to M-22.0). Two considerations argue against expecting FIM to rescue it, and against attempting a full three-way multistart comparison on it without first addressing its numerical robustness separately:

1. "Needs heavy priors, converges badly, crashes" is the signature of *many* weakly-constrained directions (the sloppy regime this note shows FIM cannot help with), not one isolated exact symmetry.
2. Crashes specifically implicate directions left *completely* unconstrained during optimization — exactly where L2's indiscriminate, every-coordinate floor is structurally better suited than FIM's single/few-direction-targeted one. Diagnosing which of possibly several overlapping near-symmetries a larger model's Hessian encodes, reliably enough to trust a projector built from it, is also a substantially harder version of the `tol`-sensitivity problem already documented for M-22.0.

The more defensible use of that model in the chapter is as a motivating, real-stakes case reported qualitatively (why a smarter regularizer was wanted in the first place), rather than as a second full quantitative comparison — unless its numerical stability is addressed as separate, scoped work first.

13 Artifact inventory

File	Purpose
`scripts/example_toy_pi3kmini_fim_fit.R`	model, one multistart per objective,
`scripts/example_toy_pi3kmini_fim_plots.R`	waterfall, group scatter, fit-vs-data,
`scripts/example_toy_pi3kmini_fim_eigenvalue_report.R`	CSV + plot of initial eigenvalue spec
`scripts/plots/pi3kmini_fim_fit/pi3kmini_results*.rds`	saved parframes/best fits/Hessians,
`scripts/plots/pi3kmini_fim_fit/pi3kmini_eigenvalue_report.csv`	eigenvalues at initial point, all ranks
`2026_PhdTex/Plots_fim/pi3kmini_cartoon.tex/.pdf`	model topology diagram

14 Open items

- Implement and test the hybrid regularizer (Section 11) across the σ sweep.

- Optional: extend PI3K-mini to two independent symmetry groups to confirm Section 10 directly (not yet run).
- Optional: a sparser/noisier variant of `ab_fit.R` to confirm Section 9's AB-model prediction directly (not yet run).
- Decide, with the above considerations in mind, whether/how to scope any use of the larger biological model in the chapter.